

What the discovery script does

A guide for whoever has to approve and run it — `EnTech-M365-Discovery.ps1` , version 1.0

This script reads your Microsoft 365 tenant and produces a report describing its size and shape, so a migration estimate rests on measured figures rather than assumptions. It exists so that nobody on your team has to spend a week counting mailboxes by hand.

In one line

It reads. It changes nothing, sends nothing, and stores no password. Output lands in a folder on the machine you run it on, and you decide whether any of it leaves.

What it does not do — and how to check each claim

Every statement below can be verified by searching the script itself. We would rather you check than take our word for it.

CLAIM	HOW TO VERIFY
It makes no change to your tenant	Search the file for <code>Invoke-MgGraphRequest</code> . There are four, and every one specifies <code>-Method GET</code> . Every Exchange cmdlet used is a <code>Get-</code> . There is no <code>Set-</code> , <code>Update-</code> or <code>Disable-</code> cmdlet in the file.
It reads no message or document content	No scope granting content access is requested — see the permission table below. The script asks for counts, sizes, names and configuration only. It could not read a message body even if it tried.
It sends nothing anywhere	Search for <code>Invoke-WebRequest</code> and <code>Invoke-RestMethod</code> — there are none. The only outbound calls are to <code>graph.microsoft.com</code> and Exchange Online, and those are the reads above. (Searching for <code>Send-</code> returns one hit: the words “Send-As permissions” inside a sentence the report prints. It is text, not a command.)
It stores no credential	Sign-in is interactive through Microsoft's own prompt. No password is typed into the script, nothing is written to disk, and the session is disconnected at the end.
The only files it writes are yours	<code>New-Item</code> creates the output folder; <code>Remove-Item</code> appears once, replacing a previous zip of the same name. Both act on local disk only.

What it collects

AREA	COLLECTED	WHY IT AFFECTS THE ESTIMATE
Tenant	Name, ID, verified domains, whether identities sync from on-premises	A domain can only live in one tenant at a time, which is what makes cutover a fixed window rather than a gradual move
Licensing	Licence types, purchased versus assigned	Shows which workloads are actually entitled, and what has to be bought in the destination
Accounts	Totals, enabled, disabled, guests, licensed, synced	Disabled and unlicensed accounts often need not move at all — frequently a large reduction
Mailboxes	Count, total and average size, item counts, the fifteen largest, archives, holds	A handful of very large mailboxes drives the schedule more than the average size does
OneDrive	Account count, storage, file counts, the largest	File count matters as much as volume — many small files migrate more slowly than few large ones
SharePoint	Site count, storage, file counts, the largest sites	Site structure and permissions are rebuilt, not copied
Teams	Team count, private channel count, visibility	Private channels each have a hidden site and are poorly handled by most tooling
Groups	Microsoft 365 groups, distribution lists, security groups	Distribution lists and their membership have to be recreated
Mail flow	Transport rules, inbound and outbound connectors, public folders	None of these migrate — each is rebuilt and tested by hand
Applications	Integrated applications, those using single sign-on, conditional access policies	The most commonly under-estimated item: every integration is re-registered and re-tested
Mail data by type	How the total splits between user mailboxes, shared mailboxes, rooms and system accounts	A tenant with far more accounts than staff usually hides most of its data outside personal mailboxes
SharePoint shape	Document libraries versus custom lists, and which sites carry them	Libraries are copied. Lists, pages and forms are rebuilt by hand — a different order of effort

Permissions it requests

Microsoft will display this list on the consent screen. Every one is a read permission; none grants access to message or file content.

PERMISSION	WHY IT IS NEEDED
<code>Organization.Read.All</code>	Tenant name, verified domains, directory-sync status
<code>User.Read.All</code>	Account counts and types — not mailbox contents
<code>Group.Read.All</code>	Groups, distribution lists, and which groups back a Team
<code>Directory.Read.All</code>	Directory-wide counts and domain configuration
<code>Sites.Read.All</code>	SharePoint site inventory — site names and properties, not documents
<code>Reports.Read.All</code>	The usage reports that carry every storage figure. This is the sizing data
<code>ReportSettings.Read.All</code>	Detects whether your reports are anonymised — see below
<code>Application.Read.All</code>	Integrated applications and SSO configuration to be rebuilt
<code>Policy.Read.All</code>	Conditional access policy count
<code>TeamSettings.Read.All</code>	Teams configuration

The Exchange Online section additionally needs the account to hold a role that can read recipient configuration — **View-Only Recipients** or **View-Only Organization Management** is enough. Global Reader covers everything above.

If your organisation does not permit this consent

Run it with `-SkipExchange`, or decline the Exchange module when prompted. The Graph sections still produce every storage figure. Alternatively, complete the questionnaire alone — we can estimate from it, with a wider range, and we will say where the uncertainty sits.

You can try it before you trust it

Run it with `-SelfTest` first. It builds a complete report from obviously fictional sample data, contacts nothing, and asks for no sign-in. That lets you see exactly what the output looks like, and confirm the script runs in your environment, before granting any permission at all. The sample report is clearly stamped as sample data.

What you need

- Windows, with **PowerShell 5.1** (built in) or **PowerShell 7**
- An account with **Global Reader**, or Global Administrator
- Internet access, and permission to install two Microsoft modules for your user account — `Microsoft.Graph.Authentication` and `ExchangeOnlineManagement`. The script asks before installing anything and will not proceed without an answer.

How to run it

1 Save the script somewhere local

For example `C:\Temp`. Avoid running it from a network share.

2 Unblock it

Windows marks files that arrived by email as untrusted, which blocks them from running. Right-click the file, choose Properties, and tick

Unblock

— or run:

```
Unblock-File -Path C:\Temp\EnTech-M365-Discovery.ps1
```

- 3 Open PowerShell and allow scripts for this session only**
This permits signed and local scripts for the current window and reverts when you close it. It changes no machine-wide setting.

```
Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass
```

4 Run it

```
cd C:\Temp  
.\EnTech-M365-Discovery.ps1
```

A Microsoft sign-in window opens. Sign in with the administrator account and review the permission list before approving.

5 Review the report, then decide what to send

It opens automatically when finished. Read it first — it contains account, site and group names from your tenant.

Useful options

```
# verify the script runs and produces a report, using sample data.  
# No tenant is contacted and no sign-in is required.  
.\EnTech-M365-Discovery.ps1 -SelfTest  
  
# skip the Exchange section entirely  
.\EnTech-M365-Discovery.ps1 -SkipExchange  
  
# use a 90-day usage window instead of 30  
.\EnTech-M365-Discovery.ps1 -ReportPeriod D90  
  
# write somewhere other than the Desktop  
.\EnTech-M365-Discovery.ps1 -OutputPath C:\Temp\Discovery
```

What it produces

- **M365-Discovery-Report.pdf** — the readable summary, EnTech-branded
- **M365-Discovery-Report.html** — the same report, if you prefer to print it yourself
- **data** — CSV extracts behind every figure, so nothing in the report is unsourced
- A **.zip** of all of the above, ready to attach to an email

It takes a few minutes for an organisation of your size. Most of that is the Teams section, which asks about each team in turn.

Things that commonly come up

The report shows hashes instead of names

Microsoft 365 can be set to conceal user, group and site names in all reports. The script detects this and says so in its output. Totals remain accurate; the detail tables become unattributable. A Global Administrator can turn it off under **Settings** → **Org settings** → **Reports**, after which re-running takes a few minutes. Turn it back on afterwards if that is your standing policy.

Sign-in is blocked by a conditional access policy

Some policies block PowerShell sign-in. Either run it from a compliant, managed device, or have an administrator add a temporary exclusion. The script cannot work around this, and should not be able to.

A section is missing from the report

That is deliberate and it is reported, not hidden. Anything the script could not collect is listed in a **Sections that could not be collected** box at the end, with the reason. A silently missing section would mean an estimate built on a gap nobody knew about.

No PDF was produced

The script converts the report using Microsoft Edge in headless mode, since Edge is present on every supported Windows build. If neither Edge nor Chrome is found, the HTML is left in place — open it in any browser and print to PDF.

If anything at all looks wrong

Stop and contact us before re-running. We would far rather answer a question than have an agency run something it is not comfortable with.

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